
Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Quantizing a large language model reduces serving cost and costs some accu-
2 racy. Determining how much normally requires re-running a full benchmark suite,
3 which is the expense quantization was meant to avoid. We ask whether the accu-
4 racy delta can be predicted instead. Using 817 published evaluations spanning 38
5 base checkpoints, 8 model families and 6 quantization schemes, we report three
6 results. First, a *negative* one: per-model prediction carries almost no signal. Under
7 leave-one-family-out validation the best learned predictor beats a global mean by
8 0.03pp of MAE, and both ridge regression and gradient boosting are *worse* than
9 guessing the average, because 33–100% of the variance on most benchmarks is
10 evaluation noise. Second, a positive one: conditioning only on the quantization
11 scheme yields split-conformal intervals with 90.1% empirical coverage (118/131,
12 95% CI [83.6, 94.6]) on checkpoints never used in construction. Third, and most
13 consequential for practice: the published corpus is outcome-truncated. Every con-
14 figuration in it is one that someone chose to release. We rented an A100 and mea-
15 sured deliberately-faulted configurations, observing losses to -39.5 pp against a
16 published worst case of -8.86 pp—measured, we stress, only on models of 1.5B
17 and below, and on configurations engineered to fail rather than typical ones. We
18 argue the resulting artifact should be described as a calibrated historical baseline
19 rather than a predictor, and we state precisely why its coverage guarantee does
20 not transfer to a practitioner’s own untuned recipe. Code, data and every figure’s
21 provenance are public.

22 1 Introduction

23 Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8
24 wants one number: how much accuracy will this cost? The honest answer today is that nobody
25 knows without running the evaluation.

26 This paper asks whether that number can be forecast from the public record. Model publishers
27 release quantized checkpoints with benchmark tables comparing the compressed model against its
28 dense parent. Aggregated, these are thousands of paired measurements of exactly the quantity of
29 interest.

30 Our contributions:

- 31 1. **A negative result** (§5). Per-model prediction of the accuracy delta does not work, and
32 we show why: the target is dominated by evaluation noise. We report this as the primary
33 finding rather than burying it.
- 34 2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured
35 coverage on held-out families and an explicit refusal rule for cells where coverage is poor.

- 36 3. **A measurement of the corpus’s own selection bias** (§7), via GPU experiments on
37 deliberately-faulted configurations—data the published record structurally cannot contain.
- 38 4. **An audit** (§9) recording every error we found in our own work, including two that changed
39 published claims.

40 2 Related Work

41 **Predicting evaluations.** BenchPress Zeng and Papailiopoulos [2026] establishes the mechanism
42 closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and
43 decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks
44 is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is a
45 model’s absolute score; ours is the paired delta between a model and its compressed self, and their
46 corpus contains no quantized checkpoints.

47 **Conformal prediction on compressed models.** Tong et al. Tong et al. [2026] apply conformal
48 prediction directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The
49 estimand differs: their conformal sets are over label space, constructed from a compressed model’s
50 own output probabilities, so the method measures uncertainty *after* running the model. Ours is con-
51 structed over historical accuracy deltas to give an interval *before* running anything. Their Figure 2
52 plots the same quantity we predict, $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$, but measures it. Their finding that com-
53 pression decouples accuracy from uncertainty bounds what an accuracy-only estimate like ours can
54 claim (§8).

55 **Quantization methods.** The checkpoints we harvest are produced with GPTQ Frantar et al. [2022]
56 and SmoothQuant Xiao et al. [2022] via llm-compressor vLLM Project [2024].

57 **The source of our data.** Kurtić et al. Kurtić et al. [2024] report over 500,000 evaluations of
58 quantized Llama-3.1 models, finding recovery above 99% of baseline. That work does not address
59 whether the published set of quantized models is a biased sample. Characterising that gap is the
60 contribution of §7.

61 **Selection bias.** Survivorship bias in benchmark corpora has been formalised and quantified in
62 information retrieval Gupta and MacAvaney [2022]. Conformal prediction’s guarantee is known to
63 fail under selection: conditional on a selection event, calibration data are no longer exchangeable
64 with a test point Barber et al. [2023], Jin and Candès [2024]. We apply this directly to our own
65 artifact in §8.

66 3 Data

67 We harvest published model cards into rows of (model, scheme, benchmark, $\text{acc}_{\text{before}}$, $\text{acc}_{\text{after}}$), yield-
68 ing 817 rows after filtering, across 38 base checkpoints, 8 families and 6 schemes.

69 **An integrity gate.** Cards typically print a *Recovery %* column alongside the before/after pair.
70 This is redundant information, and we exploit it: for every three-column row we check recovery $\approx$
71 $100 \times \text{after}/\text{before}$ to a tolerance derived from the card’s own printed precision, and reject rows that
72 disagree. This caught fabricated values arising from column-order variation between publishers, and
73 internally inconsistent cards. Rejections are recorded rather than silently dropped.

74 **A structural limitation, stated at the outset.** All rows come from a single publisher using a
75 single toolchain. Holding out a model family tests transfer across *models*; nothing in this design
76 tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

77 4 Method

78 **Split conformal.** Given calibration residuals $\{s_i\}_{i=1}^n$ and level α , the interval half-width is the
79 $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which gives exact finite-sample coverage under exchangeability.
80 Where n is too small for that index to exist, we return ∞ rather than a misleadingly finite band.

81 **Mondrian (group-conditional) calibration.** A single marginal interval advertised at 90% was
 82 in fact a 68% interval for NVFP4 users—the users who most need it. We therefore calibrate per
 83 scheme.

84 **A refusal rule.** For each (scheme, size-band) cell we measure coverage directly. Where mea-
 85 sured coverage falls below a threshold on a sufficient number of scored rows, the tool emits
 86 INSUFFICIENT CALIBRATION and no interval.

87 The refusal is scored on *one-sided* coverage: how often the true delta stayed at or above the inter-
 88 val’s lower bound. The risk this tool bounds is accuracy loss, and a model that beats its envelope
 89 has exposed no one to anything, so counting that as a failure would refuse cells that are merely out-
 90 performing. Two cells are refused: $w4a16|<2B$, where losses stayed above the lower bound 70.1%
 91 of the time over 77 rows from 2 checkpoints (two-sided containment 68.8%), and $w8a16|>10B$ at
 92 81.1% over 217 rows from 5 checkpoints (two-sided 73.7%). The distinction was identified by an
 93 external reader; the two-sided figure makes $w8a16|>10B$ look worse than the risk warrants.

94 **A support floor.** Coverage measured on many benchmarks from one checkpoint is not indepen-
 95 dent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct
 96 checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

97 5 The negative result

98 Under leave-one-family-out validation, row-weighted MAE:

	predictor	MAE (pp)	beats global mean?
	per-scheme mean (shipped)	0.7227	yes
	global mean (baseline)	0.7545	—
99	per-(scheme $\times$ benchmark)	0.7607	no
	ridge, 38 features	0.7639	no
	per-benchmark mean	0.7647	no
	gradient boosting	0.7878	no

100 Ridge and gradient boosting are both *worse* than predicting the global average. Adding model size,
 101 bit-width, method, benchmark identity and headroom all degraded out-of-family performance.

102 **Why.** Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta
 103 should be approximately zero—gives a noise floor that accounts for 33–100% of observed variance
 104 on most benchmarks. On GSM8K the noise alone is $\pm 2pp$, while the quantity being predicted has
 105 mean $-0.36pp$. The target is a difference of two noisy quantities that largely cancel: the numerically
 106 smallest and hardest-to-predict summary of quantization damage available.

107 **Implication.** The entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band,
 108 with no model and no conformal machinery, achieves comparable coverage. What conformal con-
 109 tributes is a finite-sample guarantee and a principled rule for small- n schemes, not accuracy. We
 110 therefore describe the result as a *calibrated historical baseline*, not a predictor.

111 6 Results

	scheme	90% interval (pp)	worst observed	% losing $>3pp$	evals
	fp8	[-1.11, +0.48]	-1.90pp	0.0%	80
	w8a8_int	[-1.71, +1.07]	-3.32pp	1.0%	194
112	fp8_dynamic	[-1.44, +1.34]	-8.72pp	1.0%	200
	w8a16	[-0.98, +0.81]	-3.04pp	1.2%	86
	w4a16	[-2.87, +1.40]	-8.86pp	5.7%	193
	nvfp4	[-4.10, +1.87]	-8.12pp	15.6%	64

113 **Prospective validation.** After freezing the method we harvested checkpoints not used during de-
114 velopment. Under a strict protocol that excludes any family present in training and any card whose
115 format informed a parser fix, coverage is $118/131 = 90.1\%$, 95% Clopper–Pearson CI [83.6, 94.6],
116 against a nominal 90%. A from-scratch, standard-library-only reimplementation sharing no code
117 with the pipeline reproduces this figure with zero disagreements.

118 **What the envelope cannot do.** 0 of 6 intervals exclude zero: the tool can never state that a scheme
119 *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when
120 real damage occurs, the interval usually misses it. This is the single most important caveat in the
121 paper.

122 7 The measured left tail

123 The corpus contains only configurations someone chose to publish. To measure what it structurally
124 cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

125 **Result.** Worst measured loss -39.5pp , against -8.86pp as the worst loss anywhere in the pub-
126 lished corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-
127 configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor
128 instead of per-group.

129 **Scope, stated plainly.** This tail was measured only on models of 1.5B and below. It establishes
130 that catastrophic damage is reachable; it is not a measured bound for larger models.

131 **A confound we found and did not resolve.** Re-running MMLU under the published protocol
132 reproduces published base accuracy (47.20 against 47.42; 59.57 against 60.98), closing the harness
133 question. It exposed a larger one: our own *correctly-configured* control is $2.6\times$ more damaging than
134 published W4A16 on closely comparable checkpoints (mean gap -3.78pp , $z = 2.9$ and 5.3). The
135 cause is approximately $171\times$ less calibration data than production defaults, not a bug. Consequently,
136 excess-over-control measures *bad recipe versus mediocre recipe*, not versus production. It is a
137 conservative anchor, not a bound, and we decline to publish a corrected point estimate: the correction
138 moves by 5.70pp depending on the anchor set, and we have only 6 rows from two models.

139 8 Limitations

140 **The guarantee is conditional, and void where it matters most.** Split conformal gives exact cov-
141 erage under exchangeability between calibration and test. Our calibration set is the set of configura-
142 tions a publisher chose to release—a selection event. Conditional on selection, calibration points are
143 not exchangeable with an arbitrary test point and the guarantee does not transfer Barber et al. [2023],
144 Jin and Candès [2024]. It holds for a new checkpoint drawn from the same publication process. It is
145 **void** for a recipe the reader tuned themselves and that nobody would have published—which is the
146 case a practitioner most wants covered.

147 **Accuracy is not the whole of deployment risk.** Tong et al. [2026] show compression
148 frequently decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets
149 inflate. A scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We
150 measure none of that.

151 **One publisher, one toolchain, and why.** See §3. Our own control arm (§7) is direct evidence
152 that practitioner and calibration volume can dominate scheme choice. We measured how confining
153 this is: surveying 5 publishers and 195 quantized model cards, 5 of 157 cards from the other 4
154 publishers carry paired before/after evaluations, so others do publish them. What no other publisher
155 in the sample provides is a *Recovery* column, which is what lets each row be checked against $100\times$
156 after/before and rejected on disagreement: of their rows, 0 pass that gate, against 142 from the
157 control publisher. The corpus is single-publisher not because others publish nothing, but because
158 only one publishes in a self-verifying form.

159 **Rows are not independent.** Up to sixteen benchmark rows come from a single quantization run.
160 We mitigate with group-conditional calibration and the support floor, and note that a cluster bootstrap
161 over checkpoints does not materially widen the headline interval, but the unit of analysis remains a
162 row.

163 9 Audit

164 We record errors found in our own work, including those that changed published claims. This
165 section exists because the alternative—reporting only what survived—would misrepresent how the
166 result was obtained.

167 Errors that changed a published number.

- 168 • *A retracted headline.* An early claim that held-out models came from families never used
169 in construction was wrong; they were new checkpoints of families already present. The
170 strict figure is 118/131.
- 171 • *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was
172 matched as Math-Lvl-5, contaminating 7 test rows with scores from a different bench-
173 mark.
- 174 • *A centring mismatch.* Calibration measured residuals about the stratum mean while in-
175 tervals centred on the scheme mean, mis-sizing every calibrated width. Found only by
176 independent reimplementations.
- 177 • *A column-order fault.* One publisher orders columns [recovery, base, quant], producing a
178 fabricated -68pp delta. The integrity gate rejected 10 of 11 rows on that card, passing only
179 the one where base = quant made the check degenerate.

180 **A regression we nearly shipped.** An audit indicated that a raw asymmetric empirical band cov-
181 ered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization
182 damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch.
183 On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-
184 sided empirical index requires more calibration points than are often available. An infinite interval
185 covers by construction. On rows where both are finite, the original wins on coverage *and* width. The
186 change was reversed.

187 **An overclaimed guarantee.** The artifact was described as carrying a coverage guarantee without
188 qualification. Checking the method against the conformal literature rather than against our own
189 earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected
190 in §8.

191 **Provenance discipline.** Two parser fixes were informed by inspecting rows that were already in
192 the test set. Rather than silently retain them, we record which fixes were test-informed and exclude
193 the affected family from the strict headline.

194 **Mechanical verification.** Every figure in this paper is a macro generated from a claims registry
195 that recomputes each value from source artifacts; no number here is hand-typed. The accompanying
196 repository enforces this with a test that fails if any outward-facing document contains an untraced
197 numeric literal.

198 10 Conclusion

199 Per-model prediction of quantization accuracy damage does not work on published data, and the
200 reason is measurement noise rather than model capacity. What does work is narrower: a per-scheme
201 calibrated envelope, validated at 90.1% coverage on unseen checkpoints, that refuses to answer
202 where its own measured coverage is poor. The more useful contribution may be the limitation
203 itself—the published record of quantized models is outcome-truncated, and we measured losses
204 roughly four times worse than anything it contains. A risk estimate built on that record is a floor on
205 risk, not a ceiling, and should say so.

References

- Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2), 2023.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- Prashansa Gupta and Sean MacAvaney. On survivorship bias in MS MARCO. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022. doi: 10.1145/3477495.3531832.
- Ying Jin and Emmanuel J. Candès. Confidence on the focal: Conformal prediction with selection-conditional coverage. *arXiv preprint arXiv:2403.03868*, 2024.
- Eldar Kurtić, Alexandre Marques, Mark Kurtz, and Dan Alistarh. We ran over half a million evaluations on quantized LLMs—here’s what we found. Red Hat Developer, October 2024. <https://developers.redhat.com/articles/2024/10/17/we-ran-over-half-million-evaluations-quantized-llms>.
- Yujia Tong, Yuxi Wang, Yunyang Wan, Tian Zhang, Junhao Dong, and Jingling Yuan. Does compression preserve uncertainty? a unified benchmark for quantized and sparse LLMs via conformal prediction. *arXiv preprint arXiv:2606.01850*, 2026. Wuhan University of Technology; Nanyang Technological University.
- vLLM Project. llm-compressor, 2024. <https://github.com/vllm-project/llm-compressor>.
- Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and efficient post-training quantization for large language models. *arXiv preprint arXiv:2211.10438*, 2022.
- Yuchen Zeng and Dimitris Papailiopoulos. You don’t need to run every eval. *arXiv preprint arXiv:2606.24020*, 2026. Microsoft Research, AI Frontiers.